

Cubes — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ**[1]**

The units digit of the cube of 23 is:

(a) 6 (b) 7 (c) 3 (d) 9

Q2 • MCQ**[1]**

Which of the following is a perfect cube?

(a) 100 (b) 125 (c) 150 (d) 200

Q3 • FILL IN THE BLANK**[1]**

The cube of 4 is:

Q4 • MCQ**[1]**

How many perfect cubes are there from 1 to 100 (inclusive)?

(a) 3 (b) 4 (c) 5 (d) 10

Q5 • TRUE / FALSE**[1]**

A perfect cube can end in exactly two zeros.

True False

Q6 · ASSERTION-REASON

[1]

Assertion (A): 1728 is a perfect cube.

Reason (R): $1728 = 2^6 \times 3^3$, and all its prime factors form complete groups of three.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7

[2]

Find the units digit of the cube of (i) 27 (ii) 14.

Q8

[2]

Express 27 as a sum of consecutive odd numbers.

Q9

[2]

Is 256 a perfect cube? Justify using prime factorisation.

Long answer (3 marks each)

Q10

[3]

Show, using prime factorisation, that 3375 is a perfect cube and state its cube root.

Q11

[3]

By what smallest number should 3600 be multiplied so that the product is a perfect cube? Show your working.

Q12

[3]

By what smallest number should 256 be divided so that the quotient is a perfect cube? Find the cube root of the quotient.

Total: 21 marks · [Check your work against the answer key.](#)